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DEPARTMENT OF INFORMATION TECHNOLOGY

Assignment No. 2

Subject	DBMS Lab
Assignment No.	2
Topic	DDL and DML queries
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Aim:

To study and implement DDL (Data Definition Language) and DML (Data Manipulation Language) commands using SQL for a suitable database application.

Objectives:

- To understand the concept of DDL and DML commands.
- To create database tables using DDL commands like CREATE, ALTER, and DROP.
- To manipulate data using DML commands like INSERT, UPDATE, and DELETE.
- To perform database operations on real-world applications.

Case Study: Airline Management System

SQL Queries:

Part A: DDL Commands

Theory: DDL Commands

Data Definition Language (DDL) commands are used to define and modify the structure of database objects such as tables. These commands include operations like CREATE, ALTER, and DROP. DDL commands are essential for designing the database schema.

1. Create Table: Airline

Create a table to store airline details.

```
mysql> CREATE TABLE airline (
   -> airline_id INT PRIMARY KEY,
   -> airline_name VARCHAR(50),
   -> country VARCHAR(50)
   -> );
Query OK, 0 rows affected (0.02 sec)
```

---

2. Create Table: Flight

Create a table for flight details.

```
mysql> CREATE TABLE flight (
   -> flight_id INT PRIMARY KEY,
   -> flight_name VARCHAR(20),
   -> source VARCHAR(50),
   -> destination VARCHAR(50),
   -> price DECIMAL(10,2),
   -> airline_id INT,
   -> FOREIGN KEY (airline_id) REFERENCES airline(airline_id)
```

```
    -> );
Query OK, 0 rows affected (0.03 sec)
```

---

3. Create Table: Passenger

Create a table for passenger details.

```
mysql> CREATE TABLE passenger (
    -> passenger_id INT PRIMARY KEY,
    -> pname VARCHAR(50),
    -> age INT
    -> );
Query OK, 0 rows affected (0.02 sec)
```

---

4. Alter Table

Add a new column email to passenger table.

```
mysql> ALTER TABLE passenger
    -> ADD email VARCHAR(50);
Query OK, 0 rows affected (0.01 sec)
```

---

5. Drop Table (Example)

Delete a temporary table.

```
mysql> DROP TABLE temp_data;
Query OK, 0 rows affected (0.01 sec)
```

Part B: DML Commands

Theory: DML Commands

Data Manipulation Language (DML) commands are used to manage and manipulate data stored in database tables. These commands include INSERT, UPDATE, and DELETE. DML operations help in adding, modifying, and removing records from the database.

6. Insert Data into Airline

```
mysql> INSERT INTO airline VALUES
    -> (1, 'Air India', 'India'),
    -> (2, 'Indigo', 'India'),
    -> (3, 'Emirates', 'UAE');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

---

7. Insert Data into Flight

```
mysql> INSERT INTO flight VALUES
    -> (101, 'AI101', 'Mumbai', 'Delhi', 5500.00, 1),
    -> (102, 'IN202', 'Pune', 'Bangalore', 4500.00, 2),
    -> (103, 'EM404', 'Dubai', 'Mumbai', 15000.00, 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

---

8. Insert Data into Passenger

```
mysql> INSERT INTO passenger VALUES
    -> (1, 'Rahul Sharma', 25, 'rahul@gmail.com'),
    -> (2, 'Neha Verma', 22, 'neha@gmail.com'),
    -> (3, 'Amit Patil', 30, 'amit@gmail.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

---

9. Update Query

Update passenger email.

```
mysql> UPDATE passenger
    -> SET email = 'rahul123@gmail.com'
    -> WHERE passenger_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

---

10. Delete Query

Delete a passenger record.

```
mysql> DELETE FROM passenger
    -> WHERE passenger_id = 3;
Query OK, 1 row affected (0.01 sec)
```

Case Study: Civic Voice Interface System (Mini Project)

SQL Queries:

Part A: DDL Commands

Theory: DDL Commands

DDL commands are used to create and manage the structure of database tables such as user, citizen profile, and complaint. These commands ensure proper organization of data.

1. Create Table: User

```
mysql> CREATE TABLE user (
   -> user_id INT PRIMARY KEY,
   -> name VARCHAR(50),
   -> email VARCHAR(50),
   -> phone_number VARCHAR(15)
   -> );
Query OK, 0 rows affected (0.02 sec)
```

---

2. Create Table: CitizenProfile

```
mysql> CREATE TABLE citizenprofile (
   -> profile_id INT PRIMARY KEY,
   -> user_id INT,
   -> state VARCHAR(50),
   -> district VARCHAR(50),
   -> FOREIGN KEY (user_id) REFERENCES user(user_id)
   -> );
Query OK, 0 rows affected (0.02 sec)
```

---

3. Create Table: Complaint

```
mysql> CREATE TABLE complaint (
   -> complaint_id INT PRIMARY KEY,
   -> user_id INT,
   -> issue VARCHAR(100),
   -> status VARCHAR(20),
   -> FOREIGN KEY (user_id) REFERENCES user(user_id)
   -> );
Query OK, 0 rows affected (0.02 sec)
```

---

4. Alter Table

Add language column to user.

```
mysql> ALTER TABLE user
    -> ADD language VARCHAR(20);
Query OK, 0 rows affected (0.01 sec)
```

---

5. Drop Table

```
mysql> DROP TABLE temp_complaint;
Query OK, 0 rows affected (0.01 sec)
```

Part B: DML Commands

Theory: DML Commands

DML commands are used to insert, update, and delete records in the database. These operations help in maintaining and managing real-time data efficiently.

6. Insert into User

```
mysql> INSERT INTO user VALUES
    -> (1, 'Rahul Sharma', 'rahul@email.com', '9876543210', 'English'),
    -> (2, 'Neha Verma', 'neha@email.com', '8765432109', 'Hindi'),
    -> (3, 'Amit Patil', 'amit@email.com', '7654321098', 'Marathi');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

---

7. Insert into CitizenProfile

```
mysql> INSERT INTO citizenprofile VALUES
    -> (1, 1, 'Maharashtra', 'Pune'),
    -> (2, 2, 'Delhi', 'New Delhi'),
    -> (3, 3, 'Maharashtra', 'Mumbai');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

---

8. Insert into Complaint

```
mysql> INSERT INTO complaint VALUES
    -> (1, 1, 'Road damage', 'Pending'),
    -> (2, 2, 'Water issue', 'Resolved'),
    -> (3, 3, 'Garbage issue', 'Pending');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

9. Update Query

```
mysql> UPDATE complaint
    -> SET status = 'Resolved'
    -> WHERE complaint_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

---

10. Delete Query

```
mysql> DELETE FROM complaint
    -> WHERE complaint_id = 3;
Query OK, 1 row affected (0.01 sec)
```

Conclusion:

In this assignment, DDL and DML commands were successfully implemented for both Airline Management System and Civic Voice Interface System. DDL commands helped in creating and modifying database structures, while DML commands were used to manipulate data effectively. This demonstrates the importance of SQL in managing real-world database applications.